Revenue Growth Strategy For Ford Dealers

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1. Executive Summary

Over the past few years the automobile industry has experienced dramatic transformation that has forever altered the way both new and used vehicles are sold. With increased sophistication in the automotive industry, dealers are looking for new ways to increase sales, profit margins and lower overhead costs. This report research analysis was conducted by formed by YorkU students which used a variety of analysis and models to identify vehicles and their characteristics or specifications that appeal to consumers over the last few years.

With the underlying goal of producing a model that would be able to tell dealers what type of vehicles to focus their inventories on in order to gain maximum profit, the research concluded that when it came to used cars, newer produced cars being bought by consumers yielded better profit for dealers, especially those with a bigger engine capacity, non-manual transmission type and non-petrol fuel type.

Based on the model created and analysis conducted, it is recommended that dealers move away from the ever popular manual transmission type models like the Ford Fiesta and instead focus on more automatic transmission models like the Ford Kuga. The other recommendation would be to start expanding inventory for used hybrid / electric type models for the future. Due to rising fuel costs, these used hybrid/electric vehicle options are becoming ever popular with consumers as not only do they provide financial relief with low fuel costs but they also include government tax incentives that consumers find appealing. They are also the most profitable for dealers and shifting to hybrid / electric helps Ford in its goal of becoming fully carbon neutral by 2050.

2. Introduction

Ford Motor Company is an American automaker that was founded in 1903 by Henry Ford. It is the second largest U.S- based automaker, and the fifth largest in the world. Their product line in Canada ranges from the $25,000 Ford EcoSport to the half-a-million-dollar Ford GT supercar. This report uses Ford used-cars dataset produced from 2013 to 2020, the business objective is to help Ford used-car dealers detect what types of used cars can be sold at a good price and predict the price for a car given its features. The analytical objectives are to get an overview of the dataset by studying its descriptive statistics; select independent variables for modeling; at last build models to find what factors influence car price and predict the price. To reach the final recommendation, basic analysis and data cleaning are performed on the original dataset, followed by modeling and analysis which utilize linear regression model and random forest regression model (RFR) to forecast the price for a Ford's used car.

3. Basic Analysis

This section focuses on the descriptive analysis of data features.

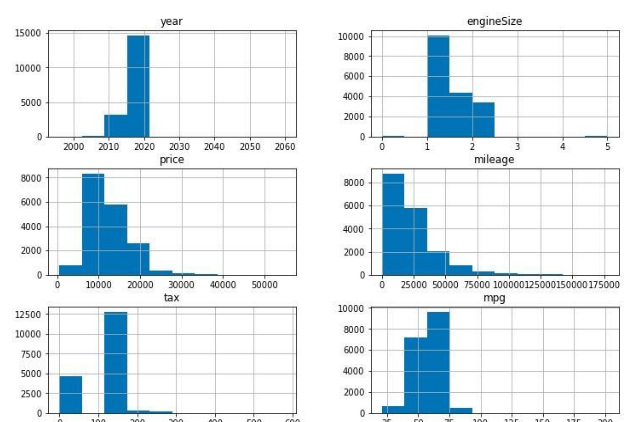
3.1 Descriptive statistics of original dataset

Below are the descriptive statistics tables and histograms for six numeric variables of the original dataset. Original dataset shows that there are 17,966 observations and each observation has full information of car features, with no missing data. In terms of six numeric features, engine size, miles per gallon (mpg) and production year have the Top 3 smallest standard deviations (0.43, 2.05 and 10.12 respectively). These three variables' means do not vary significantly from their medians. Tax, price and mileage vary the most. Meanwhile, the means of these three variables significantly vary from their medians, which will have an impact on its distribution.

Table 1

	year	engineSize	price	mileage	tax	mpg
count	17966.000000	17966.000000	17966.000000	17966.000000	17966.000000	17966.000000
mean	2016.866470	1.350807	12279.534844	23362.608761	113.329456	57.906980
std	2.050336	0.432367	4741.343657	19472.054349	62.012456	10.125696
min	1996.000000	0.000000	495.000000	1.000000	0.000000	20.800000
25%	2016.000000	1.000000	8999.000000	9987.000000	30.000000	52.300000
50%	2017.000000	1.200000	11291.000000	18242.500000	145.000000	58.900000
75%	2018.000000	1.500000	15299.000000	31060.000000	145.000000	65.700000
max	2060.000000	5.000000	54995.000000	177644.000000	580.000000	201.800000

Figure 1



In terms of three text features, the original dataset shows that a manual petrol-fueled Fiesta car is mostly sold out by customers.

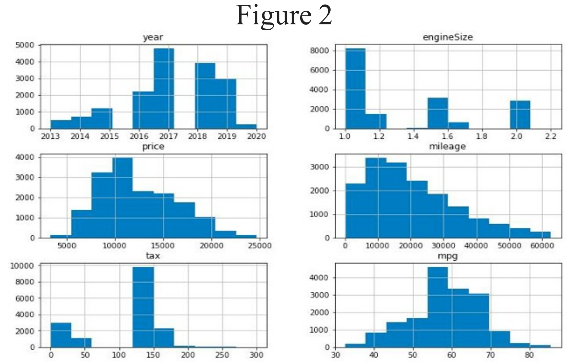
3.2 Descriptive statistics of final dataset

The skewness from above histogram indicates that there are significant outliers for the six numeric features. In order to minimize the bias in statistical models, the group dropped outliers by calculating the interquartile ranges. Below are the descriptive statistics tables and histograms for six numeric variables after the original dataset.

Table 2

	year	engineSize	price	mileage	tax	mpg
count	16448.000000	16448.000000	16448.000000	16448.000000	16448.000000	16448.000000
mean	2017.114847	1.316652	12363.672240	20336.055265	113.119042	58.242729
std	1.530264	0.375388	3957.188108	13756.226101	59.107733	8.978561
min	2013.000000	1.000000	3295.000000	1.000000	0.000000	32.500000
25%	2016.000000	1.000000	9450.000000	9825.500000	125.000000	53.300000
50%	2017.000000	1.100000	11498.000000	17375.500000	145.000000	58.900000
75%	2018.000000	1.500000	15299.250000	28860.000000	145.000000	65.700000
max	2020.000000	2.200000	24690.000000	62664.000000	300.000000	85.600000

Figure 2



Dataset after dropping outliers contains 16,448 observations, 92% of original dataset with no missing data. All the standard deviations decrease. The manual petrol-fueled Fiesta car is still mostly sold out by a customer.

4. Modeling & Analysis

This section uses two widely-used regression models for predicting a continuous dependent variable: linear regression and random forest regression (RFR), to forecast the price for a Ford used-car. Linear regression tries to find the best fitting line for input data while RFR builds multi-level decision trees and outputs the mode or mean of prediction of an individual tree.

4.1 Variable Section

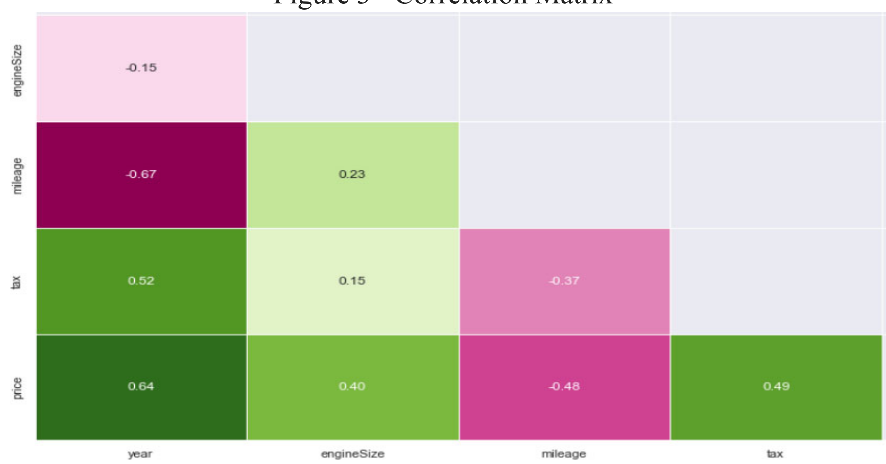
Since numeric fields are in very different scales, it is important to normalize them through dividing the values by their means and check if variables with the least variance should be dropped as they do not provide too much info.

Table 3 - Variance After Normalization

Variable	Variance	Variable	Variance
engineSize	0.081287	tax	0.273034
mileage	0.457578	**mpg**	**0.023765**

Normalized mpg's low variance in Table 3 suggests that original mpg does not have too much variation, therefore, it is dropped in the model. The simplified correlation matrix below shows that price has a high correlation with production year, a moderation correlation with engine size, mileage and tax. However, to avoid multicollinearity, the model will drop mileage as it is highly related to the year.

Figure 3 - Correlation Matrix



Finally, all the string fields are regrouped and dummy variables are created to ensure the string fields such as car model, transmission and fuel type can participate in the model.

4.2 Linear Regression

By running linear regression using 70% randomly selected training data, price can be visualized by the equation below:

$$\boxed{Price_i = 3.5year_i + 28.5tax_i + 1{,}323.7engineSize_i - 978.2Transmission_i + 941.2MD\_EcoSport_i + 2{,}111.4MD\_Focus_i + 2{,}303.7MD\_Kuga_i + 486.8MD\_Others_i + 975.1FT\_Diesel_i + 5{,}525.2FT\_Others_i}$$

where everything except year, tax and engine size is a dummy variable and reference car is a petrol Fiesta car. The coefficients indicate that engine size, transmission, car model and fuel type all have important influences on price. For example, holding everything else constantly, a manual car is $978 cheaper than other transmission type cars. A Fiesta car is at least $486 cheaper than other models. Hybrid and electricity-fueled cars are $5,525 more expensive than a petrol car. By entering the features of a car, this model can predict its price.

The good side of this model is all the factors' influence can be quantified. However, the score shows that this model can only explain 41.5% of the price's variance. The root of mean squared error (RMSE) of both train and test data is above 3,000, meaning that the average gap between actual and predicted price is over $3,000 after penalizing the outliers. Last but not least, the year's impact on price is not as crucial as what is suggested in the correlation matrix. Although it is hard to find reasons for these limitations as all the assumptions for linear regression are met, the group will try the other model and see if the fitting score can be improved.

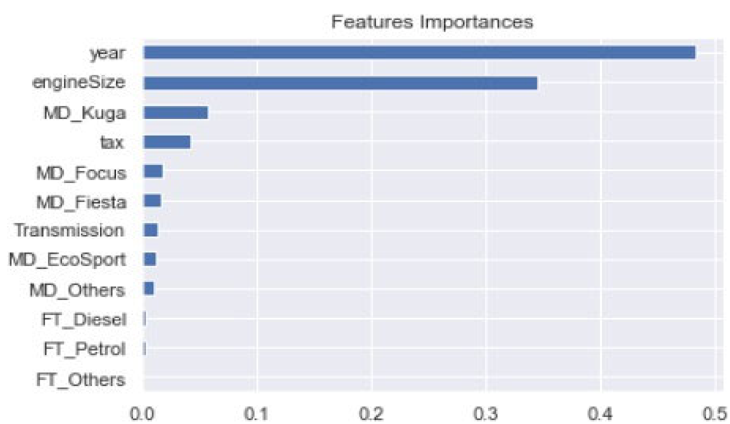
Table 4 - Model Validation For Linear Regression

Model Score	RMSE: Training Data	RMSE: Test Data
0.415	3,007.79	3,066.71

4.3 Random Forest Regression

To improve model fitting score, RFR is deployed. The horizontal bar chart below shows the production year, engine size, Kuga car and tax are the Top 4 important features in deciding a used-car's price.

Figure 4 - Feature Importance In Random Forest Regression



Compared to linear regression, RFR's model score is significantly improved to 89% as this model is a combination of learning models that increases the overall result (Donges, 2022). RMSE of both datasets also decrease significantly, meaning that the average gap between predicted and actual prices is narrowed and the prediction is more accurate.

Table 5 - Model Validation For Random Forest Regression

Model Score	RMSE: Training Data	RMSE: Test Data
0.890	1,304.19	1,389.13

By entering the features of a car, this model can predict its price. However, the limitation of RFR is that the impact of each feature cannot be quantified. Meanwhile, it will lose effectiveness when a car's features are not covered in the sample data range. For example, holding everything else constantly, a car produced in 2025 will report the same price as a 2020 car (Rebellion, 2022).

Below charts show how predicted prices vary from actual prices in two models.

Figure 6 - Predicted vs Actual Price: Linear

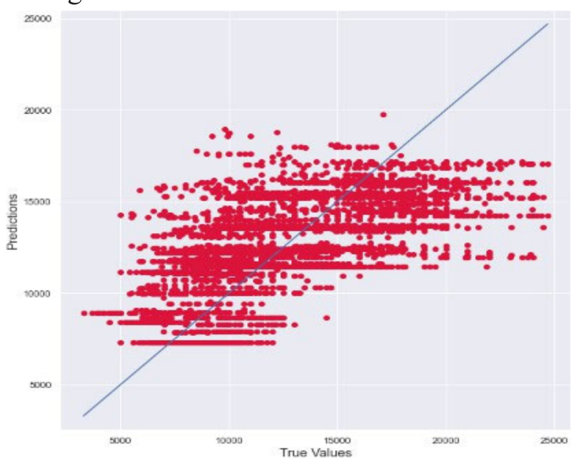
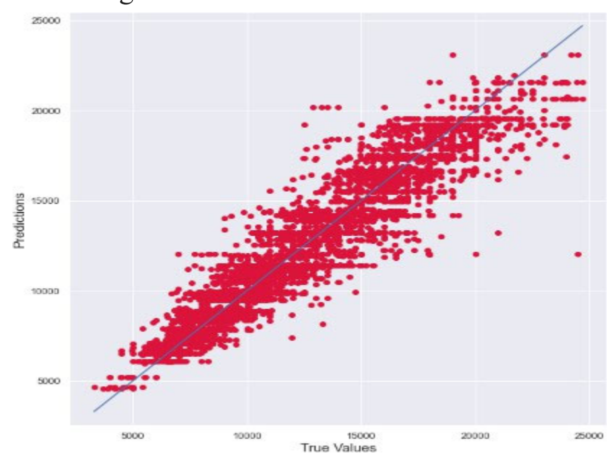


Figure 7 - Predicted vs Actual Price: RFR



One thing to note is that since the training and test dataset are randomly selected, the coefficients, fitting scores, RMSE will change whenever running the regression for a new time. However, the rank of feature importance, high RFR score while lower linear regression score, and high RMSE for linear regression while lower RMSE for RFR, do not change.

5. Conclusion

In conclusion, 16,448 Ford used-cars dataset were used out of the 17,966 observations in the original dataset. Based on the results from the linear regression model and random forest regression model (RFR), two recommendations are made to the client. First is to get a good price for a Ford used car, it is important to focus on the cars that are newly produced, with a higher engine size. Also, try to avoid traditional cars such as manual-petrol cars, especially shift from Fiesta to Kuga and Focus and try to find more hybrid, electric cars. Lastly, RFR model is suggested when clients need a prediction for price of the cars, because it gives a more accurate predicted price where it's closer to the actual price.

6. Reference

Donges, N. (2022, July 6). *Random Forest Algorithm: A Complete Guide*. Built in. https://builtin.com/data-science/random-forest-algorithm

Rebellion. (2022, April 7). *What Are The Disadvantages Of Random Forest?* Rebellion https://www.rebellionresearch.com/what-are-the-disadvantages-of-random-forest